

Seminars 2

	R_1	R_2	R_3
R_1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$
R_2	0	$\frac{1}{2}$	$\frac{1}{4}$
R_3	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$

Example 2.3.3

$$1 - \frac{1}{2} = \frac{1}{2} \quad : 2 \quad \Rightarrow \frac{1}{4}$$

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$$

$$\text{Lumi } \Delta^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{Monti: } P \cdot \Delta^{(0)} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix} = \Delta^{(1)}$$

$$\text{Miorari } P \cdot \Delta^{(1)} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{3}{8} \\ \frac{1}{8} \\ \frac{1}{2} \end{pmatrix} = \Delta^{(2)}$$

$$\text{Jo } P \cdot \Delta^{(2)} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{3}{8} \\ \frac{1}{8} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{11}{32} \\ \frac{3}{16} \\ \frac{15}{32} \end{pmatrix} \rightarrow \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$\text{Dupa } k\text{-zile } P^k \begin{pmatrix} \Delta_1^0 \\ \vdots \\ \Delta_m^0 \end{pmatrix}$$

$$\Delta = P \Delta$$

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{pmatrix} = \begin{pmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} - 1 & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{2} - 1 & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{2} - 1 \end{pmatrix} \begin{pmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 1 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{4} & 1 & 0 \\ \frac{1}{2} & \frac{1}{4} & -\frac{1}{2} & 1 & 0 \end{pmatrix} \xrightarrow{L_3+L_1} \begin{pmatrix} -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 1 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{4} & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{4} & 1 & 0 \end{pmatrix} \xrightarrow{L_3+L_2}$$

$$\sim \begin{pmatrix} -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 1 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{4} & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\Rightarrow \Delta_3 = t$$

$$-\frac{1}{2}\Delta_2 + \frac{1}{4}\Delta_3 = 0 \quad | \cdot 4$$

$$-2\Delta_2 + \Delta_3 = 0 \quad \Rightarrow \Delta_2 = \frac{t}{2}$$

$$-\frac{1}{2}\Delta_1 + \frac{1}{4} \cdot \frac{t}{2} + \frac{1}{4}t = 0$$

$$\Rightarrow \Delta_1 = \frac{3}{4}t$$

$$\Rightarrow \Delta = \begin{pmatrix} \frac{3t}{4} \\ \frac{t}{2} \\ t \end{pmatrix}$$

$$\frac{3t}{4} + \frac{t}{2} + t = 1 \quad \Rightarrow \quad t = \frac{4}{9}$$

Pe termen lung: probabilitățile sunt următoarele:

$\frac{1}{3}$ vorneau în R_1

$\frac{2}{9}$ vorneau în R_2

$\frac{4}{9}$ vorneau în R_3

Culegere

↓
Markov pag 138
140

Exercițiul 2.9.4 (sus, mijloc, jos)

$$P = \begin{matrix} & \begin{matrix} S & M & J \end{matrix} \\ \begin{matrix} S \\ M \\ J \end{matrix} & \begin{pmatrix} 0,7 & 0,1 & 0,1 \\ 0,1 & 0,8 & 0,3 \\ 0,2 & 0,1 & 0,6 \end{pmatrix} \end{matrix}$$

$$a) \quad \Delta^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \text{pt lower-class}$$

$$\Delta^{(1)} = P \cdot \Delta^{(0)} = \begin{pmatrix} 0,1 \\ 0,3 \\ 0,6 \end{pmatrix}$$

$$\Delta^{(2)} = P \cdot \Delta^{(1)} = \dots = \begin{pmatrix} 0,16 \\ 0,43 \\ 0,41 \end{pmatrix}$$

b) $\Delta = P\Delta$

$$\Leftrightarrow \begin{pmatrix} -0,3 & 0,1 & 0,1 & | & 0 \\ 0,1 & -0,2 & 0,3 & | & 0 \\ 0,2 & 0,1 & -0,4 & | & 0 \end{pmatrix} \xrightarrow[L_1 \sim L_2]{L_1 \leftrightarrow L_2} \begin{pmatrix} 0,1 & -0,2 & 0,3 & | & 0 \\ -0,3 & 0,1 & 0,1 & | & 0 \\ 0,2 & 0,1 & -0,4 & | & 0 \end{pmatrix} \cdot 10$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & | & 0 \\ -3 & 1 & 1 & | & 0 \\ 2 & 1 & -4 & | & 0 \end{pmatrix} \xrightarrow[L_2 - 2L_1]{L_2 + 3L_1} \begin{pmatrix} 1 & -2 & 3 & | & 0 \\ \dots & \dots & \dots & | & \dots \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 3 & | & 0 \\ 0 & -5 & 10 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$$\Delta_3 = t$$

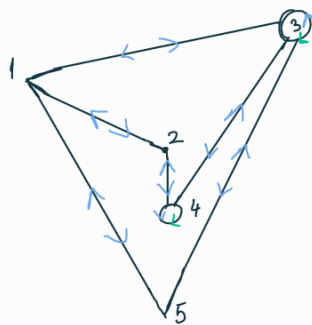
$$-\Delta_2 + 2\Delta_3 = 0 \Rightarrow \Delta_2 = 2t$$

$$\Delta_1 - 2\Delta_2 + 3\Delta_3 = 0 \Rightarrow \Delta_1 - 4t + 3t = 0 \Rightarrow \Delta_1 = -t$$

$$t + 2t + t = 1 \Rightarrow t = \frac{1}{4} \Rightarrow \Delta = \begin{pmatrix} \frac{1}{4} \\ \frac{1}{2} \\ \frac{1}{4} \end{pmatrix}$$

Exercițiul 2.9.8

Dacă vom cu 2 iuriri
colomne lui 3 orote oze



$$\begin{pmatrix} \frac{1}{5} \\ 0 \\ \frac{2}{5} \\ \frac{1}{5} \\ \frac{1}{5} \end{pmatrix}$$

	1	2	3	4	5
1	0	$\frac{1}{2}$	$\frac{1}{4}$	0	$\frac{1}{2}$
2	$\frac{1}{3}$	0	0	$\frac{1}{3}$	0
3	$\frac{1}{3}$	0	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$
4	0	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{3}$	0
5	$\frac{1}{3}$	0	$\frac{1}{4}$	0	0

poate prim 4 drumuri
unul care iese din
3 si se intoarce in 3

$$\Delta^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Delta^{(2)} = P \cdot \Delta^{(1)} = \begin{pmatrix} \frac{5}{12} \\ 0 \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{12} \end{pmatrix}$$

$$\Delta^{(1)} = P \Delta^{(0)} = \begin{pmatrix} 0 \\ \frac{1}{5} \\ \frac{1}{3} \\ 0 \\ \frac{1}{3} \end{pmatrix}$$

$$\Delta^{(3)} = P \cdot \Delta^{(2)} = \dots$$

$$P\Delta = \Delta$$

Ex. 2.9.12

$$P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}, \quad 0 < p < 1, \quad 0 < q < 1$$

a) $\frac{1}{p+q} \begin{bmatrix} q \\ p \end{bmatrix}$ vectorul $P \cdot \Delta = \Delta$

$$P \cdot \Delta = \Delta, \quad \Delta = \begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix}$$

$$\begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix} \begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix} = \begin{bmatrix} \Delta_1 \\ \Delta_2 \end{bmatrix}$$

$$\begin{pmatrix} 1-p-1 & q \\ p & 1-q-1 \end{pmatrix} = \begin{pmatrix} 0 & q \\ p & 0 \end{pmatrix} \quad \begin{pmatrix} -p & q \\ p & -q \end{pmatrix} = \begin{pmatrix} 0 & q \\ p & 0 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} -p & q & 0 \\ p & -q & 0 \end{array} \right) \xrightarrow{L_2 + L_1} \left(\begin{array}{cc|c} -p & q & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$\Delta_2 = t \Rightarrow -p\Delta_1 + tq = 0 \Rightarrow -p\Delta_1 = -tq$$
$$\Delta_1 = \frac{tq}{p}$$

$$\Rightarrow \Delta = \begin{pmatrix} \frac{q}{p+q} \\ \frac{p}{p+q} \end{pmatrix} = \frac{1}{p+q} \begin{pmatrix} q \\ p \end{pmatrix}$$

$$P^m = \frac{1}{p+q} \begin{bmatrix} q & q \\ p & p \end{bmatrix} + \frac{(1-p-q)^m}{p+q} \begin{bmatrix} p & -q \\ -p & q \end{bmatrix}$$

Dem prin inducție

$$\text{I } m=1 \quad P = \frac{1}{p+q} \begin{bmatrix} q & q \\ p & p \end{bmatrix} + \frac{1-p-q}{p+q} \begin{bmatrix} p & -q \\ -p & q \end{bmatrix}$$

$$= \begin{pmatrix} \frac{q}{p+q} + \frac{p(1-p-q)}{p+q} & \frac{q}{p+q} - \frac{q(1-p-q)}{p+q} \\ \frac{p}{p+q} - \frac{p(1-p-q)}{p+q} & \frac{p}{p+q} + \frac{q(1-p-q)}{p+q} \end{pmatrix}$$

$$= \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix} = P$$

$$\mathbb{I} \quad P \cdot P^k = \frac{1}{p+q} \begin{bmatrix} q & q \\ p & p \end{bmatrix} + \frac{(1-p-q)^k}{p+q} \begin{bmatrix} p & -q \\ -p & q \end{bmatrix} \quad (A)$$

$$P^{k+1} = P^k \cdot P = \frac{1}{p+q} \begin{bmatrix} q & q \\ p & p \end{bmatrix} \cdot \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix} + \frac{(1-p-q)^k}{p+q} \begin{pmatrix} p & -q \\ -p & q \end{pmatrix} \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix}$$

$$= \dots \quad \text{Adoro}$$

$$\lim_{m \rightarrow \infty} (1-p-q)^m = 0 \quad \rightarrow \in (-1, 1)$$

$$\begin{array}{l} 0 < p < 1 \\ 0 < q < 1 \\ \hline 0 < p+q < 2 \end{array} \quad / -1$$

$$-1 < p+q-1 < 1 \quad / (-1)$$

$$-1 < 1-p-q < 1$$